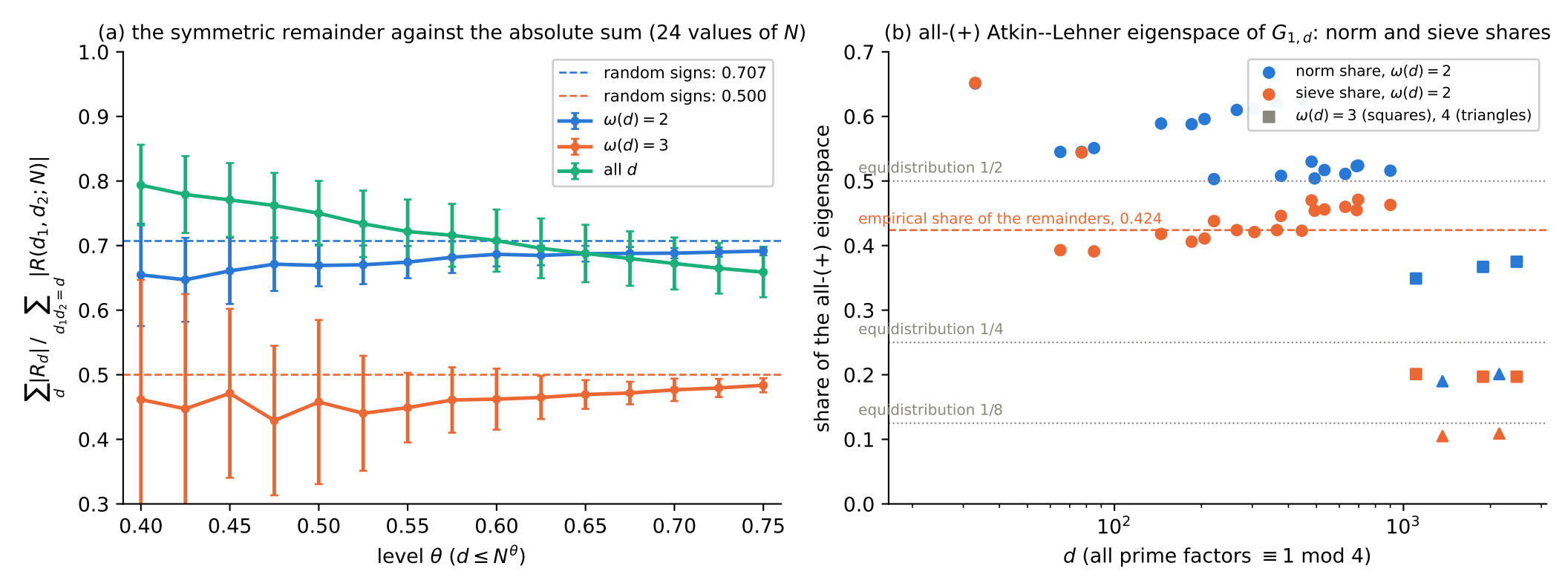
(a) the symmetric remainder against the absolute sum (24 values of N)
random signs: 0.707
random signs: 0.500
$\omega(d) = 2$
$\omega(d) = 3$
all d
$\sum_d |R_d| \,/\, \sum_{d_1 d_2 = d} |R(d_1, d_2; N)|$
level θ ($d \le N^\theta$)
(b) all-(+) Atkin--Lehner eigenspace of $G_{1,d}$: norm and sieve shares
norm share, $\omega(d) = 2$
sieve share, $\omega(d) = 2$
$\omega(d) = 3$ (squares), 4 (triangles)
equidistribution 1/2
empirical share of the remainders, 0.424
equidistribution 1/4
equidistribution 1/8
share of the all-(+) eigenspace
d (all prime factors $\equiv 1$ mod 4)